SUSTAINABILITY



The Future of Fuel Cells in Data Centers

- High Temperature Fuel Cells -

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Panel Discussion



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Agenda

1. High Temp Fuel (HTFC) Cells: Gigawatts deployed, still a new technology for data centers

Market context, public acceptance

2. Chemistries, Use Cases, Readiness

Solid Oxide Fuel Cells (SOFC), Molten Carbonate Fuel Cells (MCFC)

3. Benefits and Challenges

Power, Combined Cooling and Power (CCHP), Carbon Capture, Long Duration Energy Storage (LDES)

4. Comparison with incumbent solutions

Efficiency, Footprint, Cost Curves

5. Hub Acceleration & OCP collaboration

Activity Tracks and Scaling Initiatives

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High Temperature Fuel Cells: GW Deployed and still a New Technology for Data Centers

1.5-2 GW of SOFC capacity deployed worldwide to date
0.35 to 0.50 GW MCFC deployed Globally

PROOF POINT · FLAGSHIP SITE

Oracle — Project Jupiter (New Mexico)

Oracle replaced planned gas turbines/diesel generators with a SOFC microgrid.

2.45 GW
Planned capacity — one of the largest data center microgrids in the US

92%
reduction in NOx emissions vs. the original turbine-based design

Source: Oracle newsroom; Data Center Dynamics, Apr 2026

PROOF POINT · UTILITY OFFTAKE

AEP — Regulated Utility Commitment

American Electric Power signed a 20-year offtake agreement for SOFC capacity — the first major regulated-utility bet on fuel cells at scale.

1 GW
contracted SOFC capacity

$2.65B
20-year offtake agreement, finalized Jan 2026

Source: EnkiAI market tracking, 2026

PROOF POINT · MULTI-SITE SCALE

Equinix — Distributed Portfolio Rollout

Equinix expanded its agreement to deploy SOFCs across a growing portfolio — proof fuel cells scale beyond a single flagship site.

100+ MW
of SOFC capacity under contract

19+
data centers across the Equinix portfolio

Source: Equinix Energy announcements; EnkiAI market tracking, 2026

https://nachhaltigwirtschaften.at/en/iea/technologyprogrammes/afc/ ; https://www.scribd.com/document/600606629/TheFuelCellIndustryReview2021-4

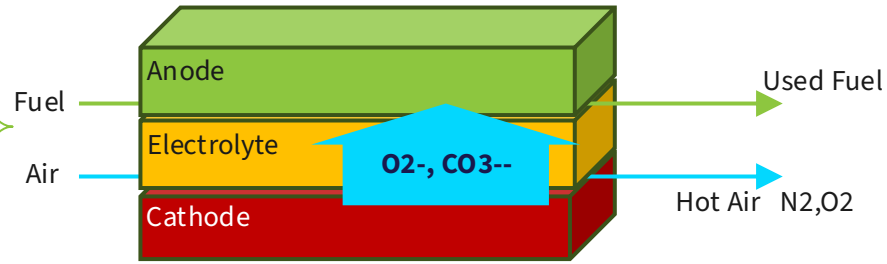
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Technology & Chemistries

- ***Solid Oxide Fuel Cells (SOFC):*** *SOFC represents roughly 60% to 66% of all stationary fuel cell megawatts installed globally*
- ***Molten Carbonate Fuel Cells (MCFC):*** *MCFC accounts for roughly 15% to 18% of cumulative stationary installations.*

SOFC is the HT-chemistry most frequently used for data centers, and it will be the focus of this presentation

	Chemistry	Operating Temperature
High Temperature	**SOFC (Solid Oxide)**	**500–900°C**
	MCFC (Molten Carbonate)	**~550°C**
Low Temperature	PEMFC (Proton Exchange Membrane)	60–80°C
	PAFC (Phosphoric Acid)	~150–200°C
	AFC (room-temp)	23–70°C



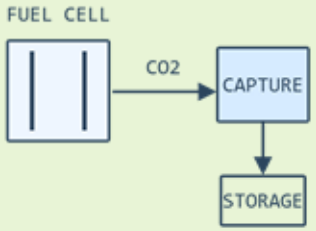
No mixing of air and fuel and no combustion in the stack:
The electrolyte acts as an Air Separation Unit.

Why is this relevant ?
The used fuel output consists of **highly concentrated CO2** (when the fuel contains carbon).

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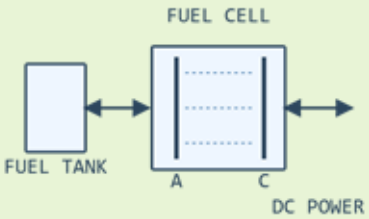
Use Cases of HT Fuel Cells

Highest-value pilot targets precisely because they aren't proven at scale yet — a first-mover reference architecture here carries outsized industry influence.



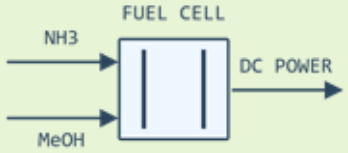
"Gas without Emissions" Affordable and scalable Carbon Capture

Natural-gas-fed SOFC/MCFC + carbon capture & storage (CCS) — firm power approaching net-zero, bridging to green hydrogen



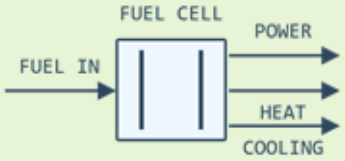
HT-Fuel Cells as LDES

HT-FC can store energy in clean molecules at high efficiency



Fuel Flexibility Pilot with Clean Fuels: NH3, MeOH

Clean fuels are strategic options for store and use clean energy. NH3 and MeOH are potential fuels for marine applications, with established supply chains



Combined Power and Cooling

Trigeneration: extend CHP heat reuse into absorption cooling for the data hall, improving effective PUE

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Readiness Analysis: Technology and Adoption T&ARL

TRL By use cases	TRL **6** Subsystem Prototype in Relevant environment	TRL **7** System Prototype in Relevant environment	TRL **8** Final System in Relevant environment	TRL **9** Commercial System in real environment
	Reversible, LDES	Power Gen /alternative fuels, **CCS** **Power-to-Fuels, Electrolysis**		**Power Generation** with Nat Gas

ARL for Data Centers Applications ARL Category	**SOFC (Solid Oxide)**	**MCFC (Molten Carbonate)**
Value Proposition	**High Cost**	**High Cost**
Market readiness		
Resource maturity	**Few mature suppliers**	**Few mature suppliers**
License to Operate		

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Benefits & Advantages of High Temp Fuel Cells

EFFICIENCY & REVERSIBILITY

High Efficiency

>60%+ from kW to GW nominal and partial loads.

HT-FC Hybrid Cycles

Can reach ultra-high 90%+ electrical efficiency.

Reversible Design

Seamless Power-to-Fuel & Fuel-to-Power capabilities for LDES.

SYSTEM & INTEGRATION

Modularity & Redundancy

Inherent redundancy allows elimination of backup generators.

«DC Native» Architecture

Efficient integration with modern 800 V architectures.

Zero Combustion

Virtually zero NOx/SOx emissions and quiet operation (65dB @10ft) driving higher public acceptance.

SPECIFIC HT-FC BENEFITS

High-Grade Heat

Co-generation of high-grade heat, power, and cooling streams.

Fuel Flexibility

Adaptable to clean fuels including Hydrogen (H_2), Biogas (CH_4), Methanol(MeOH), and Ammonia (NH_3).

Carbon Capture

Enables the lowest cost carbon capture integration.

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Operational Challenges of High Temp Fuel Cells

THERMAL STRESS & STARTUP

Extreme Operating Temperatures

Slow Startup Times: Requires prolonged heat-up and cool-down periods (operating at 600°C–1000°C), restricting rapid cyclic flexibility.

Material Degradation: Extreme thermal cycling accelerates mechanical stress, chemical incompatibility, and structural breakdown.

VERIFY / VALIDATE

Elimination of back-up generators

Load following capabilities

Durability data

HIGH INITIAL CAPITAL COST

Financial Barriers to Entry

Premium Capex: Initial equipment and deployment costs remain substantially higher than traditional diesel or gas combustion engines.

High Stack Cost: Expensive specialized materials and complex fabrication processes keep stack and manufacturing expenses elevated.

VERIFY / VALIDATE

Cost reduction and total installed costs

Stack materials and manufacturing

COMPLEX MAINTENANCE

Operational Rigor & Lifespan

Scheduled maintenance: Inherent system complexity mandates strict maintenance routines and planned service intervals.

Degradation Rates: Typical efficiency loss of 0.2% to 2.0% per 1,000 hours requires stack refurbishment within 5–10 years.

VERIFY / VALIDATE

Actual downtime & Maintenance intervals

VERIFY LCOE: installed capex & Opex: fuel and maintenance

Efficiency Comparison

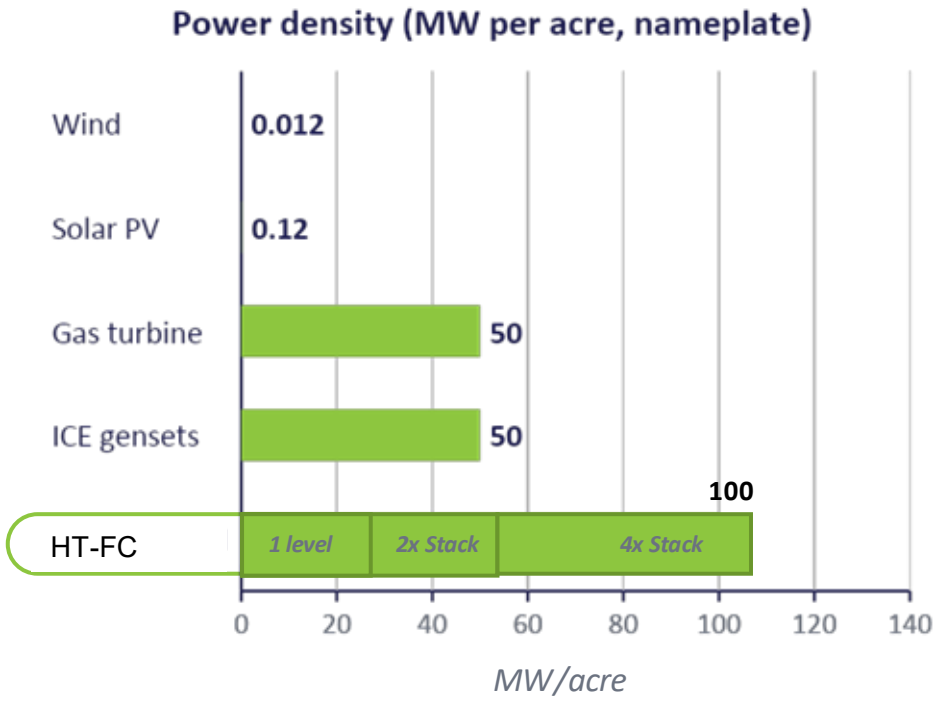
Small Aeroderivative (Simple Cycle) vs. Large Combined-Cycle Gas Turbines vs. SOFC Fuel Cells — Sources: Major Suppliers

Technology	Capacity per Unit	Efficiency	Efficiency Claim
Small Aeroderivative Gas Turbine (Simple Cycle — Data Centers)	20–50 MW	**35–42%**	
Large Combined-Cycle Gas Turbine (Gigawatt-Scale Data Centers)	**300–600 MW** (GT); up to 640 MW (1x1 combined-cycle)	**Up to 64%**	
SOFC Fuel Cell	**~0.3 MW** (300–330 kW) per module	**65%** Beginning of Life 53% cumulative	**53–65%** cumulative electrical efficiency (LHV, net AC), with an industry-leading average lifetime electrical efficiency of ~54%; >90% total efficiency when heat is recovered (CHP).
	~0.3 MW (300–330 kW) per module	**~60%**	**~60%** electrical efficiency demonstrated on **100% hydrogen**

Footprint Compared: Power Density by Generation Technology

How much power per acre? A key advantage for space-constrained data center campuses.

Power Generation Technology	MW/acre	Basis / Notes
Wind (utility-scale)	**~0.012**	~3 MW/km² nameplate, onshore spacing
Solar PV (utility-scale)	**~0.12**	~25–37 W/m² nameplate, utility-scale ground-mount
Gas turbine plant (whole plant, incl. ancillaries, open cycle)	**~50**	Up to ~50 MW/acre for gas turbines, data center siting basis
ICE generators (diesel/gas gensets)	**~50**	Up to ~50 MW/acre for reciprocating engines, incl. fuel storage
Fuel cells — SOFC	**25-100**	25 1 level, 50 2xstack, 100 4xStack

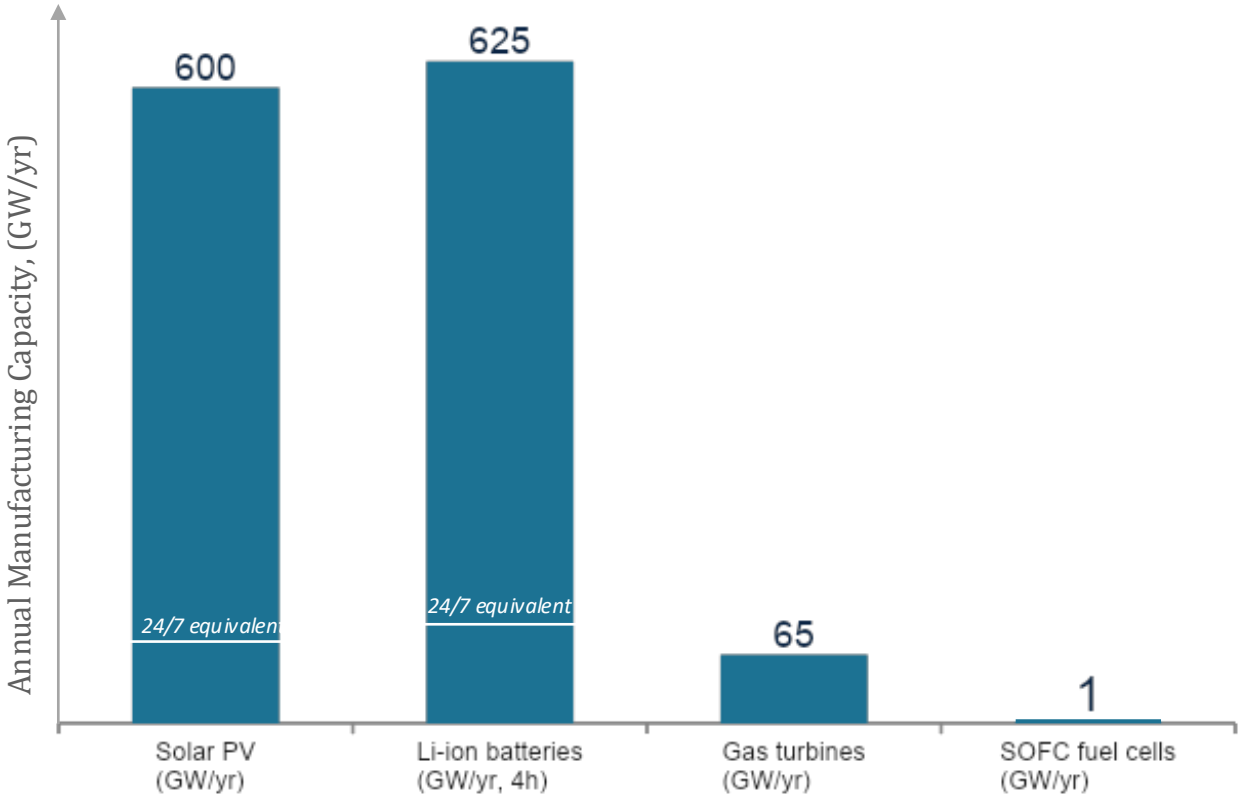


Sources:Wind/solar nameplate density: Turkovska et al., Cell Reports Sustainability, 2024 sciencedirect.com/science/article/pii/S2590332224004755 | Environ. Res.: Infrastruct. Sustain., 2026 iopscience.iop.org/article/10.1088/2634-4505/ae437f 1 acre = 4,046.86 m². | CoreSite (turbines/recips up to 50 MW/acre) coresite.com/blog/fuel-cells-onsite-power-generation-solution-for-data-centersBloom Energy / AEP, 14 Nov 2024 (100 MW per acre, 4-high stacked) bloomenergy.com/news/bloom-energy-announces-gigawatt-fuel-cell-procurement-agreement-with-aep-to-power-ai-data-centers

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Cost-Down Potential with Capacity Build-Out

Increase of Annual Manufacturing Capacity drives → **Cost Decline Since Early Commercial Deployment**



Solar PV

~99%+ decline since 1977

(Swanson's Law: ~20% per doubling of volume)

Li-ion Batteries

~90% decline since 2010

SOFC

~50–65% decline since 2010

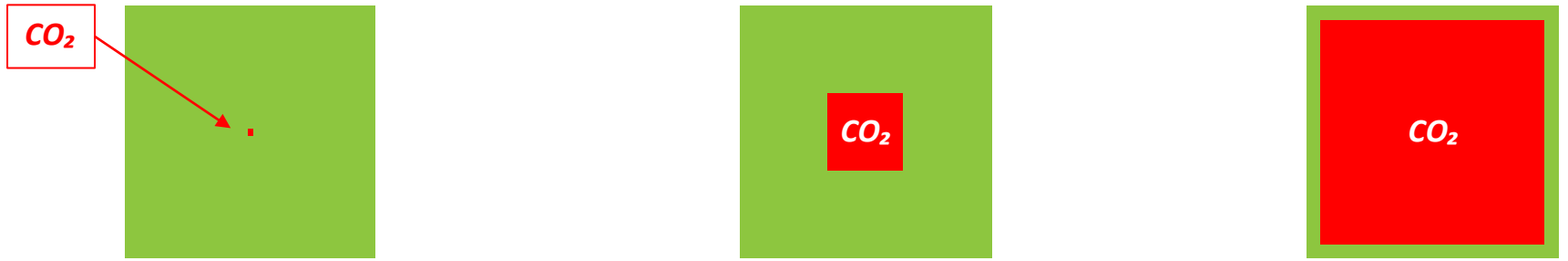
Substantial cost down potential with increasing manufacturing capacity

Note: all bars expressed as GW/yr of power capacity. Li-ion battery capacity converted from ~2,500 GWh/yr of cell output assuming a 4-hour discharge duration (2,500 GWh ÷ 4 h ≈ 625 GW).

Sources: Fraunhofer ISE, Avenston solar cost history; IEA Batteries and Secure Energy Transitions 2025; Thunder Said Energy / Utility Dive Bloom Energy cost tracking

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Carbon Capture with High Temperature Fuel cells



	Direct Air Capture (DAC)	Combustion generators (gas turbine / ICE)	High-temperature fuel cell (SOFC)
CO_2 concentration at source	**~420 ppm (0.04%)**	**~4–14% CO_2**	**~30–80% CO_2 (anode exhaust, dry basis)**
Why it matters for capture	Extreme dilution — large fans must process huge air volumes; highest capture energy & cost per ton CO_2.	Fuel is burned in air (~80% N2) — flue gas diluted by atmospheric N_2; needs post-combustion amine scrubbing to concentrate CO_2.	Fuel and air streams stay physically separate — anode exhaust is inherently CO_2-rich, simplifying capture.

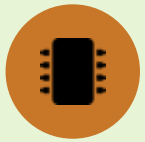
Takeaway: Because **HTFCs** generate power electrochemically rather than by combustion, fuel is never diluted with atmospheric nitrogen. The result is an anode exhaust stream with a far higher, more concentrated CO_2 content. making carbon capture integration simpler and less energy-intensive to bolt onto a fuel cell plant.
MCFC can simultaneously extract CO2 from the exhaust of combustion based systems, removing the own emission and the emissions of existing plants.

Sources: ScienceDirect, solid oxide fuel cells with integrated direct air carbon capture, techno-economic study (2024); USPTO fuel cell anode-exhaust CO_2 capture filings; DXP, "Carbon Capture vs. Direct Air Capture"; Chalmers University, SOFC power plants with integrated CO_2 capture.

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The Hub Approach to de-risk and accelerate HTFC deployment

SOFC application development, validation, and completing the path to bankability all require large-scale, long-duration testing — which requires funding.
The Hub's mission is to de-risk technology, demand, regulation and funding collectively, involving innovators and investors, rather than leaving each hyperscaler to solve it alone.



Technical Alignment

Joint industry requirements for technology developers — specifically optimizing fuel cell integration into 800V DC architectures



Market Signaling

A unified, industry-wide market signal attract investments and accelerates technology maturity



Supply Chain Evaluation

Cultivate and validate multiple supply chains, reducing supply-chain risks.



Validation at scale of HTFC application and products

Systematically evaluate and benchmark reversible-SOFC LDES applications, Carbon Capture, Combined Cooling and power, Use of alternative fuel such as direct ammonia.

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Requested Input by Track

Chemistry Status and potential Track	Validate TRL and ARL scores per chemistry; finalize/expand company landscape table
Applications Track	Confirm application status matrix; scope CC / combined cooling / CDR pilot proposals; confirm Microsoft CDR alignment
Design Standards Track	Confirm OCP document references; provide single-line integration diagram; define electrical interface parameters

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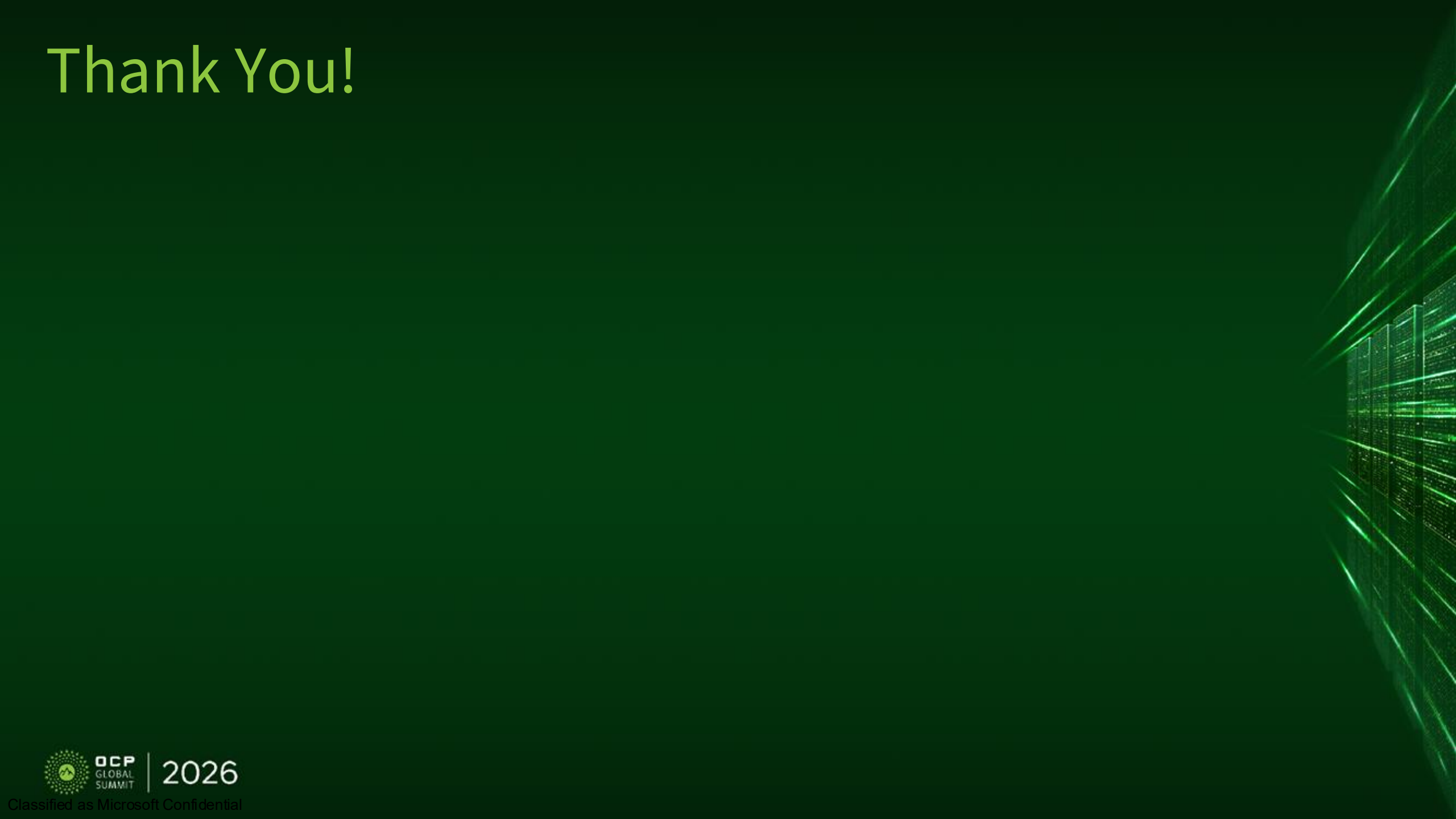
Call to Action

- Problem to solve: increase deployment of a proven innovative technology.
  - Start an initiative on HT-Fuel Cells for data centers in the Energy sub-project
    - Develop a specification for HT-Fuel Cells integration in Data Centers, design standards and interfaces
    - Validate innovative applications of HT-Fuel Cells in Data Centers (pilots) through collaboration with the NZIH
- How to get involved in the project/sub-project community
  - Reach out to Alberto Ravagni
- Timeline for contribution availability
  - 2027 OCP Regional Conference



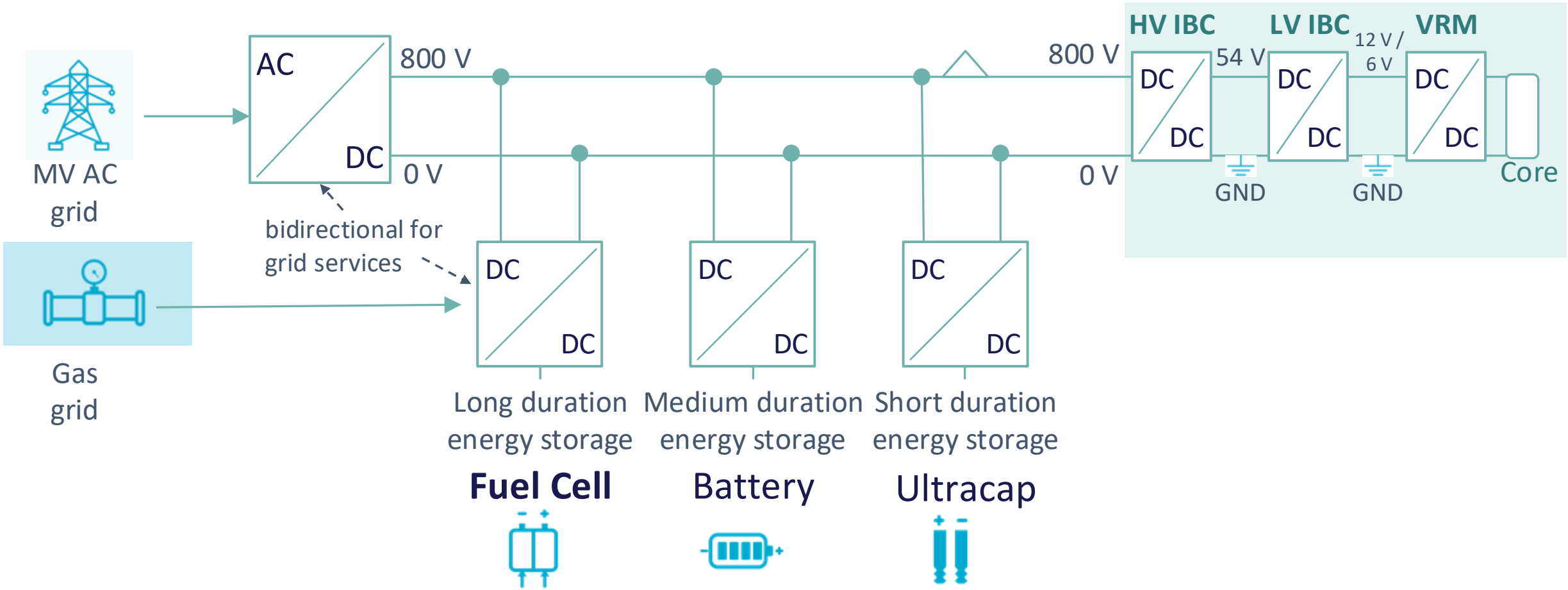
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Thank You!



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Integration of “gas batteries” in 800V DC Architecture



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Sources & Reference Documents

- Original Hub proposal email: “Collaborative OCP Initiative: High-Temperature Fuel Cells (SOFC) for Data Centers,” A. Ravagni, 31 May 2026
- Session abstract source doc: “High Temperature Fuel Cell Technologies_OCP.docx”
- OCP Foundation, “Power Architecture Evolution in Data Centers,” white paper v1.0, Sept 25 2025 — opencompute.org
- OCP Community — Power Distribution workstream; DCF Power Distribution LVDC white paper v1.0 — opencompute.org
- NVIDIA, “Building the 800VDC Ecosystem for Efficient, Scalable AI Factories” — developer.nvidia.com
- Schneider Electric, “5 Principles for 800VDC in AI Data Centers” (SPD_WP213_EN)
- Oracle Newsroom — Project Jupiter / BorderPlex / Bloom Energy announcement, Apr 2026; Data Center Dynamics
- Utility Dive — fuel cell noise/emissions vs. gas turbines for data center siting
- Bloom Energy corporate history — Wikipedia, company press materials; FuelCell Energy corporate history — Zippia, company materials
- Noon Energy / Meta partnership announcement — noon.energy
- Equinix/Vertiv “EcoEdge PrimePower (E2P2)” consortium announcements — investor.equinix.com, vertiv.com
- Fraunhofer ISE Photovoltaics Report; Avenston solar PV cost history; IEA “Batteries and Secure Energy Transitions” 2025
- Thunder Said Energy / Utility Dive — Bloom Energy cost-per-kW tracking
- Microsoft carbon removal / direct air capture waste-heat pilot — Data Center Dynamics

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Operational Challenges of High Temp Fuel Cells

THERMAL STRESS & STARTUP

Extreme Operating Temperatures

Slow Startup Times: Requires prolonged heat-up and cool-down periods (operating at 600°C–1000°C), restricting rapid cyclic flexibility.

Material Degradation: Extreme thermal cycling accelerates mechanical stress, chemical incompatibility, and structural breakdown.

HIGH INITIAL CAPITAL COST

Financial Barriers to Entry

Premium Capex: Initial equipment and deployment costs remain substantially higher than traditional diesel or gas combustion engines.

High Stack Cost: Expensive specialized materials and complex fabrication processes keep stack and manufacturing expenses elevated.

COMPLEX MAINTENANCE

Operational Rigor & Lifespan

Scheduled Downtime: Inherent system complexity mandates strict maintenance routines and planned service intervals.

Degradation Rates: Typical efficiency loss of 0.2% to 2.0% per 1,000 hours requires stack refurbishment within 5–10 years.

Maintenance Intervals Examples

Reciprocating Engines vs. Gas Turbines vs. SOFC Fuel Cells

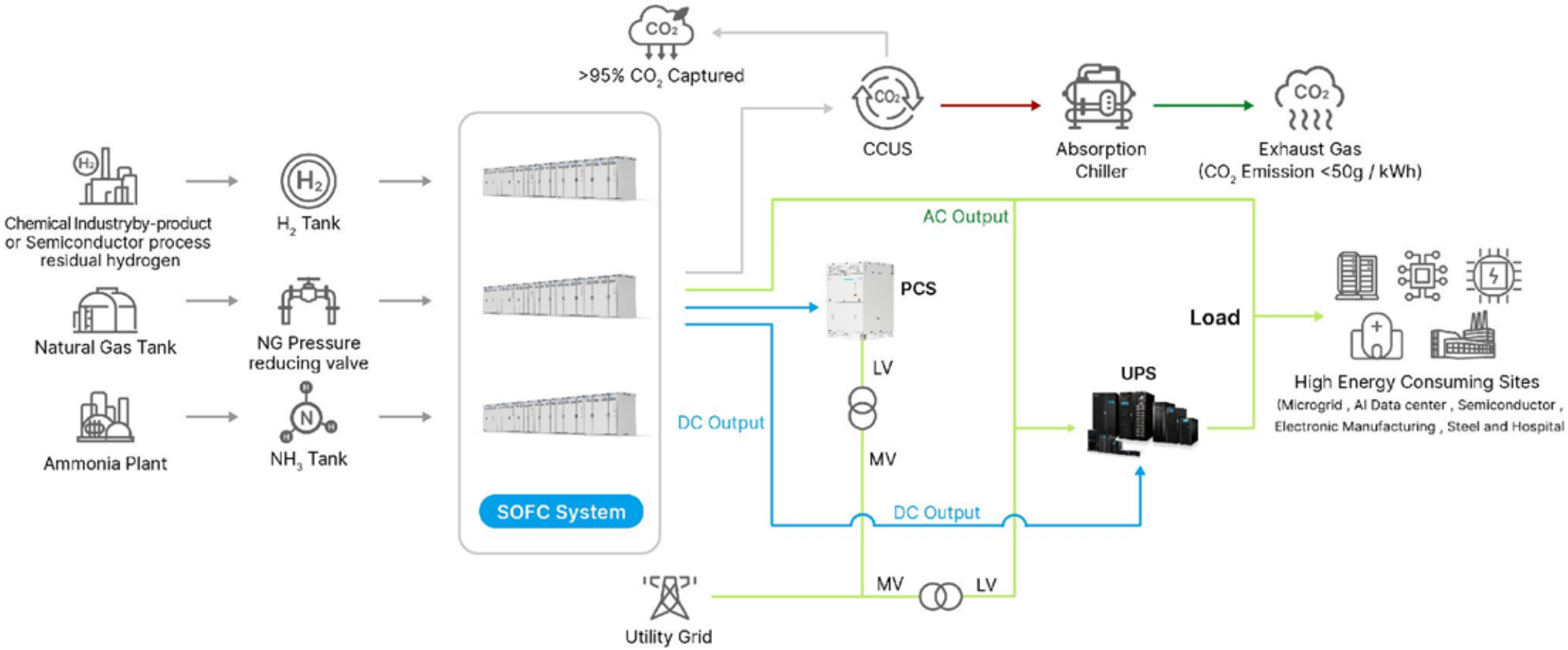
Technology	Interval	Statement / Claim	References
Reciprocating Engine	Every 2,000 hrs	Spark plugs, oil filter, valve clearances checked/replaced.	GE Jenbacher gas genset maintenance schedule
	Every 4,000 hrs	Charge air cooler, air filter, crankshaft alignment checked.	Depco — Natural Gas Generator Maintenance
	Every 8,000 hrs	Prechamber and ignition system serviced.	Caterpillar — Gas-Fueled Genset White Paper
	Every 12,000 hrs	Turbocharger and waste gate inspected/serviced.	PowerGen Enterprises — Maintenance Schedule
	Every 16,000 hrs	Cylinder heads, liners, bearings (con rod & main), piston rings — top-end overhaul (24–36 service hrs; interval depends on OEM B10/B50 life basis).	Caterpillar — Gas-Fueled Genset White Paper
	Every 24,000 hrs	HT and LT water pumps, starter replaced/serviced.	Depco — Diesel Generator Maintenance Checklist
	Every 32,000 hrs	Gear train, camshaft bearing, exhaust manifold — major/in-frame overhaul (~2× top-end interval).	PowerGen Enterprises — Maintenance Schedule
	Every 64,000 hrs	Replace all major components (full major overhaul).	Depco — Diesel Generator Maintenance Checklist
Gas Turbine	Every 4,000 hrs	Borescope inspection (1 day), no disassembly required.	Turbine Parts Specialist — LM2500 Checklist
	Every 25,000 hrs	Hot-section / combustor / rotor exchange (outage 1–2 weeks).	Turbine Parts Specialist — LM2500 Checklist OxMaint — Hot Gas Path Guide
	Every 50,000 hrs	Core exchange / major overhaul (outage 1 week–1 month; combustion inspections ~10 days vs. up to 10 weeks for full overhaul).	Turbine Parts Specialist — LM2500 Checklist Fox Innovation & Technologies
SOFC Fuel Cells	Every 45,000 hrs	Hot swap of fuel cell stack, on-site— no system downtime. (Vendor-published figure.)	Bloom Energy — Field Longevity & Reliability

Performance Criteria Comparison

Gas Turbines vs. SOFC Fuel Cells — Ambient Temperature, Altitude & Part-Load Efficiency

Technology	Parameter	Statement / Claim	References
Gas Turbine	**Ambient Temperature**	Power output falls ~5–10% per 10°C rise above 15°C ISO (up to ~26% loss at 45°C); empirical studies show ~0.1% efficiency and ~1.47 MW loss per 1K rise; rule of thumb ~0.4% power loss per 1°F above 59°F.	Wärtsilä — Derating due to Ambient Temperature Applied Thermal Engineering — De Sa & Al Zubaidy (2011) Power Engineering — Keeping Cool Keeps Output High
	Altitude	Power output and efficiency decrease by roughly 3–3.5% for every 1,000 ft (305 m) of elevation gain above sea level.	Gas Turbine Performance Analysis (citing Petchers, 2002)
	Part-Load Efficiency	**Degrades sharply at partial load** — simple-cycle efficiency drops to ~30% at 50% load; modeling shows a 5–8 percentage-point efficiency loss per 20% load reduction.	Wärtsilä — Part-Load Efficiency Comparison Off-Design GT Performance (TechScience, 2020)
SOFC Fuel Cell	**Ambient Temperature**	Output modeled/marketed as **essentially flat** (~100%) across ambient temperature range. Independent literature support is strongest for SOFC–gas turbine hybrid systems, not standalone units— flagged as vendor-sourced for standalone SOFC.	DOE OSTI — SOFC–GT Hybrid System Study (2018)
	Altitude	Output modeled/marketed as **near-flat with only gradual decline** (~97–98% at 5,000 ft) in vendor performance materials.	Genesal Energy — Derating due to Altitude & Temperature
	Part-Load Efficiency	Stays comparatively flat across load range; hybrid SOFC systems show **flat efficiency between 75–85% fuel utilization**, with slight improvement at lower current density.	DOE OSTI — SOFC–GT Hybrid System Study (2018) ScienceDirect Topics — Part Load Performance

Overview HTFC integration



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